

MICKAËL FRIBOULET

Embedded Systems Cybersecurity Student

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Driving license B

21 y.o

PROFESSIONAL EXPERIENCE

M1 Internship – IoT Offensive Security (LoRaWAN)

Serma Safety & Security - Rennes

📅 Apr 2025 – Aug 2025

- Attacks on LoRaWAN: **replay**, **eavesdropping**, **bitflipping**, etc.
- Wrote technical training sheets (theory and practical) for an IoT security toolkit.
- Used specific tools and hardware: HackRF One, WHAD, LoRAtack, nRF, Raspberry Pi, etc.

ACADEMIC PROJECTS

LoRaWAN Protocol Attacks Reproduction

University of South Brittany

📅 Oct 2024 – Feb 2025

- Simulated vulnerabilities in IoT radio protocols.

3-Axis Stabilizer Design (Steadycam)

University of South Brittany

📅 Nov 2024

- Real-time embedded control using FreeRTOS.

CSAW ESC 2024 European Finals

University of South Brittany

📅 Nov 2024

- Finalist at the *Embedded Security Challenge* (CSAW ESC), Europe.

GPS Transmission via LoRa

University of South Brittany

📅 Apr 2024 – May 2024

- Designed a long-range embedded system for GPS coordinate transmission using LoRa.

EDUCATION

Master's in Embedded Systems Cybersecurity (CSSE)

University of South Brittany

📅 Sep 2024 – Present

📍 Lorient, France

Bachelor in Digital Systems, Embedded Computing and IoT (SNIO)

University of South Brittany

📅 Sep 2021 – Jun 2024

📍 Lorient, France

ABOUT ME

Master's student in Embedded Systems Cybersecurity at the University of South Brittany, passionate about vulnerability analysis in IoT and wireless communications (LoRaWAN, NB-IoT, etc.), and applied cryptography for embedded devices. Currently seeking an internship from **January 2026** for a minimum of **12 weeks**, in the above-mentioned fields.

SOFT SKILLS

Rigor, analytical thinking, autonomy, curiosity, teamwork

SKILLS

Embedded Systems:

STM32, nRF, FreeRTOS, Zephyr OS, UART, SPI, I2C

Radio & SDR:

LoRa / LoRaWAN, HackRF One, GNU Radio Companion, WHAD, LoRAtack

Offensive Security:

Eavesdropping, replay, bitflipping attacks; reverse engineering (Ghidra); key extraction, Wireshark, GDB

Cryptography:

AES, RSA, Round Robin, embedded cryptography, Secure Boot

Programming:

Python, C, Bash, Java, Assembly, VHDL

Tools & Environments:

Nordic SDK, nRF Connect, STM32CubeIDE, STM32CubeProgrammer, Markdown, LaTeX

Electronics:

Analog and digital electronics

Operating Systems:

Linux, Windows

LANGUAGES

- **French:** Native
- **English:** Intermediate (B2)
- **Spanish:** Intermediate (B1)